A factor-two obstruction to the global-phase Jacobian variational problem

Candidate full-solution packet for arXiv:1611.00653

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Status: candidate full negative answer, likely valid. The Schwartz-class variational construction requested in the Discussion of Section 7.2 cannot exist. For every measurable set E, its proposed Jacobian integral is at most one half of the total Dirichlet energy.

1 The question

Carbonaro and Dragičević ask whether there is a measurable $E \subset \mathbb{R}^2$, with both E and E^c of positive measure, such that for every $\delta > 0$ one can find real functions $\rho > 0$ and ω with $u = \rho e^{i\omega} \in \mathcal{S}(\mathbb{R}^2)$ and

$$\int_{\mathbb{R}^2} |\nabla u|^2 < (1+\delta) \int_E \mathscr{J}(\rho^2, \omega). \tag{1}$$

Here $\mathscr{J}(F, G) = F_x G_y - F_y G_x$. This is the final variational question in the Discussion of Section 7.2 of the source paper [1].

The answer is negative, independently of the choice of E.

Theorem 1 (Universal factor-two obstruction)**.** *Let $E \subset \mathbb{R}^2$ be measurable. Suppose $\rho > 0$ and ω are real and sufficiently regular for the displayed expressions below, and suppose $u = \rho e^{i\omega} \in \mathcal{S}(\mathbb{R}^2)$. Then*

$$\boxed{2 \int_E \mathscr{J}(\rho^2, \omega) \le \int_{\mathbb{R}^2} |\nabla(\rho e^{i\omega})|^2.} \tag{2}$$

Consequently (1) is impossible for every $0 < \delta \le 1$. In particular, no set E has the property requested in the source paper.

2 Proof

Put

$$J = \mathscr{J}(\rho^2, \omega).$$

The polar decomposition gives the exact energy identity

$$|\nabla u|^2 = |\nabla \rho|^2 + \rho^2 |\nabla \omega|^2. \tag{3}$$

In particular $\rho \in L^2$, $\nabla\rho \in L^2$, and $\rho\nabla\omega \in L^2$. Since

$$J = 2\rho(\rho_x \omega_y - \rho_y \omega_x),$$

Cauchy–Schwarz and $2ab \le a^2 + b^2$ give the pointwise estimate

$$|J| \le 2|\nabla\rho|\, \rho|\nabla\omega| \le |\nabla\rho|^2 + \rho^2|\nabla\omega|^2 = |\nabla u|^2. \tag{4}$$

Thus $J \in L^1(\mathbb{R}^2)$.

The global real phase is now decisive. Define

$$V = (\rho^2 \omega_y, -\rho^2 \omega_x).$$

Then, distributionally,

$$\operatorname{div} V = J. \tag{5}$$

Moreover

$$\|V\|_{L^1} \le \|\rho\|_{L^2}\,\|\rho\nabla\omega\|_{L^2} < \infty.$$

Let χ_R be a standard cutoff which is one on the ball B_R, vanishes outside B_{2R}, and satisfies $|\nabla\chi_R| \le C/R$. From (5),

$$\left|\int_{\mathbb{R}^2} \chi_R J\right| = \left|\int_{\mathbb{R}^2} \nabla\chi_R \cdot V\right| \le \frac{C}{R}\|V\|_{L^1} \longrightarrow 0.$$

Dominated convergence, using $J \in L^1$, yields

$$\int_{\mathbb{R}^2} J = 0. \tag{6}$$

Write $J_+ = \max\{J, 0\}$ and $J_- = \max\{-J, 0\}$. Equation (6) implies

$$\int J_+ = \int J_- = \frac{1}{2}\int |J|.$$

For every measurable E we therefore have, by (4),

$$\int_E J \le \int J_+ = \frac{1}{2}\int |J| \le \frac{1}{2}\int |\nabla u|^2,$$

which is (2).

Finally, if (1) held for some $0 < \delta \le 1$, its right-hand side would be positive and (2) would give

$$\int |\nabla u|^2 < (1+\delta)\int_E J \le \frac{1+\delta}{2}\int |\nabla u|^2 \le \int |\nabla u|^2,$$

a contradiction. This proves the theorem. $\square$

3 Why the obstruction was hidden

Pointwise near-equality in (4) is exactly the Cauchy–Riemann mechanism suggested in the source. Such a region creates positive oriented Jacobian. But a globally defined phase and Schwartz decay force the total signed Jacobian to vanish by (6). Every unit of positive Jacobian mass must therefore be balanced by an equal amount of negative mass elsewhere, and (4) charges the energy of both. This doubles the lower bound and rules out constants approaching one.

The conclusion answers the displayed Schwartz-class problem exactly. It does not by itself characterize semigroup contractivity for every rough matrix field, nor does it exclude variants in which the phase is not global or the domain has a boundary carrying nonzero Jacobian flux.

4 Novelty check and review note

Cheap run indexes contained no result for arXiv:1611.00653. Bounded searches through 11 August 2026 using the exact variational wording, the paper title, and the terms "Schwartz", "Jacobian", and "global phase" found the source paper but no later answer to this question. This is not an exhaustive novelty claim. The mathematical proof uses only the polar energy identity, an L^1 divergence cutoff, and the positive/negative-part decomposition; each integrability step is explicit above.

References

[1] A. Carbonaro and O. Dragičević, *Convexity of power functions and bilinear embedding for divergence-form operators with complex coefficients*, J. Eur. Math. Soc. 22 (2020), 3175–3221, doi:10.4171/JEMS/984; arXiv:1611.00653.